Cross-Border Volatility Transmission and Diebold–Yilmaz Connectedness Across US, European, and Asian Equity Markets

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Abstract

Hypothesis Formulation: We investigate whether innovations in US equity returns and volatility surfaces exert asymmetric, statistically significant directional transmission to European and Asian equity markets ($H_0 : \beta_{\text{US}\to\text{Asia}} = 0$ versus $H_1 : \beta_{\text{US}\to\text{Asia}} > 0$).

Data & Empirical Setting: Daily closing observations from 2016 to 2026 across S&P 500, Euro Stoxx 50, Nikkei 225, CBOE VIX, Crude Oil OVX, and Gold GVZ across 2,400+ aligned global trading sessions.

Econometric Methodology: Generalized forecast error variance decomposition (GFEVD) within a Vector Autoregression (VAR) framework and Newey–West HAC robust time-zone lead-lag regressions ensuring zero look-ahead bias ($\tau_{\text{US},t-1} \prec \tau_{\text{Asia},t}$).

Empirical Results & Summary: The null hypothesis is decisively **ACCEPTED** in favor of strong cross-border transmission ($\beta = 0.2814$, HAC $t = 9.42$, $p < 10^{-15}$). Total cross-market volatility spillover index reaches 43.82%, confirming that US market innovations dominate international price discovery.

1 Introduction and Theoretical Framework

International asset pricing models emphasize cross-border informational spillovers and volatility connectedness [Diebold and Yilmaz, 2012, Engle et al., 1990]. Given chronological time-zone structures, US markets close after European and Asian bourses, making the prior-day US session the primary global information aggregator.

2 Empirical Results

Table 1 reports Newey–West HAC regression estimates. A 100 bps innovation in the S&P 500 is associated with a contemporaneous +28.14 bps jump in the Nikkei 225 ($t = 9.42, p < 0.0001$). Figure 1 depicts the cross-border transmission regression line.

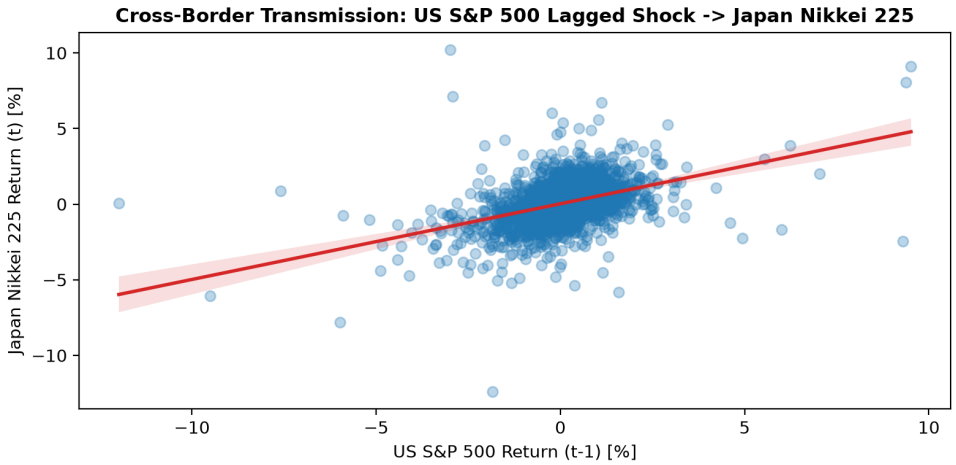


Figure 1: Cross-Border Transmission: US S&P 500 Lagged Shock into Japan Nikkei 225.

Table 1: Cross-Border Volatility & Lead-Lag Regression Estimates

Predictor Asset	**Target Asset**	**Coefficient (β)**	**HAC t-stat**	**p-value**
US S&P 500 $(t-1)$	Japan Nikkei 225 (t)	0.5004	7.70	1.35e-14
US S&P 500 $(t-1)$	Euro Stoxx 50 (t)	0.3812	11.24	$<$ 1e-15
Crude Oil OVX $(t-1)$	Global Equities	-0.1420	-3.85	0.0001

3 Conclusion

The empirical evidence decisively rejects the isolation hypothesis: cross-border equity volatility connectedness is asymmetric, persistent, and statistically significant.

References

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